LOCALLY FINITELY PRESENTED AND FINITELY PRESENTED MORPHISMS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 4158-4163 and proof span 4165-4181 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement is a two-sentence assertion without numbered parts; the proof separates the local statement from the additional quasi-compact and quasi-separated conditions, as the author does.

Editorial note. 5 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The verification that being of finite presentation is a local property of ring maps, carried out in the same chapter's characterization of locally finitely presented morphisms. Cited at proof lines 4165-4167. Allows the locality machinery for ring map properties to be applied to finite presentation.
- The transfer of a local property of ring maps to the corresponding property of morphisms, cited from the same chapter. Cited at proof lines 4167-4169. Turns the algebraic stability statement into the geometric one.
- Base change of a finite presentation ring map is of finite presentation, cited from the algebra chapter. Cited at proof lines 4169-4171. Supplies the algebraic input for the first statement.
- Quasi-compactness and quasi-separatedness are preserved by base change, cited from the schemes chapter. Cited at proof lines 4172-4177. Upgrades the first statement to morphisms of finite presentation, which combine local finite presentation with quasi-compactness and quasi-separatedness.

Source reference. Original lemma 05G5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7049–7049. Same-chapter reference used in the retained author text at source line(s) 2700, 2745, 4123, 4173. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers

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Source reference. Original lemma 01SV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2352–2352. Same-chapter reference used in the retained author text at source line(s) 2720, 3908, 4144, 4169, 5832, 5847, 7112, 7127, 7711, 8129, 8144. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01TQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2383–2383. Same-chapter reference used in the retained author text at source line(s) 4141, 4166, 4288, 4818, 5230, 5274, 5335, 5525, 6717. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01K5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2038–2038. Same-chapter reference used in the retained author text at source line(s) 1319, 2335, 2748, 3954, 4177, 4356. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01KU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2063–2063. Same-chapter reference used in the retained author text at source line(s) 4153, 4178, 8851, 9912, 9926. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *The base change of a morphism which is locally of finite presentation is locally of finite presentation. The same is true for morphisms of finite presentation.*

Proof. In the proof of Lemma 01TQ we saw that being of finite presentation is a local property of ring maps. Hence the first statement of the lemma follows from Lemma 01SV combined with the fact that being of finite presentation is a property of ring maps that is stable under base change, see Algebra, Lemma 05G5. By the above and the fact that a base change of a quasi-compact, quasi-separated morphism is quasi-compact and quasi-separated, see Schemes, Lemmas 01K5 and 01KU we see that the base change of a morphism of finite presentation is a morphism of finite presentation. □